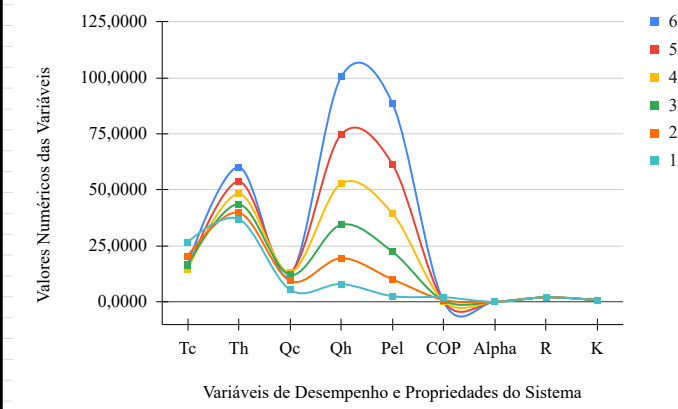


Cenário Nominal (Brisa Marítima Moderada) (Sistema de Refrigeração a Ar)									
I	Tc	Th	Qc	Qh	Pel	COP	Alpha	R	K
6	16,7066	60,1181	11,8482	100,4724	88,6242	0,1337	0,0490	2,1073	0,8161
5	14,5197	53,6857	13,2646	74,7429	61,4784	0,2158	0,0493	2,0728	0,8109
4	14,4487	48,2017	13,3106	52,8068	39,4962	0,3370	0,0495	2,0506	0,8076
3	16,4328	43,6082	12,0255	34,4326	22,4071	0,5367	0,0496	2,0402	0,8060
2	20,4725	39,8750	9,4091	19,4999	10,0908	0,9325	0,0496	2,0414	0,8062
1	26,6245	36,9981	5,4246	7,9925	2,5679	2,1125	0,0495	2,0545	0,8082

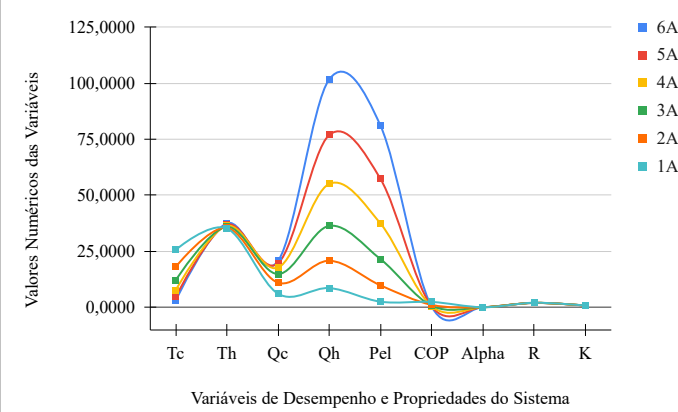
I	dTh/dI	dQc/dI	dCOP/dI	dTc/dI	dTc/dL_ins
6	6,9334	-2,1256	-0,0690	3,2819	-44,40
5	5,9461	-0,7211	-0,0977	1,1134	-44,40
4	5,0315	0,6226	-0,1507	-0,9613	-44,40
3	4,1605	1,9476	-0,2650	-3,0070	-44,40
2	3,3065	3,2914	-0,5918	-5,0818	-44,40
1	2,4435	4,6896	-2,3564	-7,2407	-44,40



```
if __name__ == "__main__":
    params = {
        'dim_bloco': (0.4, 0.27, 0.23),
        'l_ins': 0.04,
        'k_ins': 0.035,
        'h_int': 8.0,
        'h_ext': 22.0,
        'T_amb': 35.0,
        'n_TEC': 1,
        'alpha0': 0.050,
        'R0': 2.0,
        'K0': 0.8,
        'beta_alpha': -0.0015,
        'beta_R': 0.004,
        'beta_K': 0.0015,
        'cooling_type': 'air',
        'R_hs': 0.25,
        'm_dot_w': 0.05,
        'cp_w': 4180.0,
        'T_w_in': 35.0,
        'R_block': 0.02
    }
    sys = ParallelThermoelectricSensitivity(params)
```

Cenário Nominal (Brisa Marítima Moderada)(Sistema de Refrigeração a Watercooler)									
I	Tc	Th	Qc	Qh	Pel	COP	Alpha	R	K
6	3,0195	37,5225	20,7130	101,7753	81,0624	0,2555	0,0504	1,9622	0,7943
5	4,6418	36,9068	19,6623	76,9345	57,2722	0,3433	0,0503	1,9662	0,7949
4	7,6371	36,3659	17,7223	55,1101	37,3878	0,4740	0,0502	1,9760	0,7964
3	12,0716	35,9011	14,8502	36,3570	21,5069	0,6905	0,0501	1,9919	0,7988
2	18,0328	35,5153	10,9892	20,7896	9,8004	1,1213	0,0499	2,0142	0,8021
1	25,6310	35,2128	6,0681	8,5866	2,5186	2,4093	0,0496	2,0434	0,8065

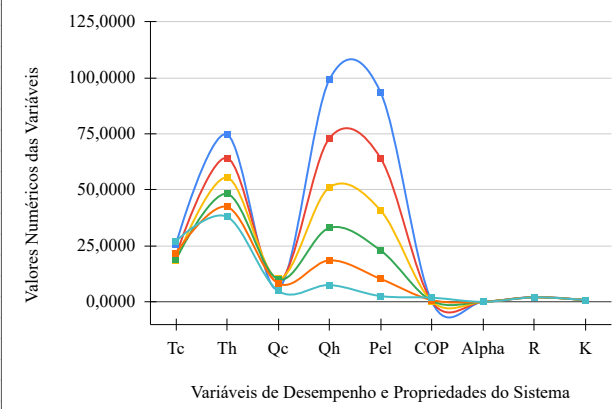
I	dTh/dI	dQc/dI	dCOP/dI	dTc/dI	dTc/dL_ins
6	0,6529	0,6169	-0,0735	-0,9524	-45,5900
5	0,5784	1,4894	-0,1050	-2,2996	-45,5900
4	0,5032	2,3978	-0,1629	-3,7021	-45,5900
3	0,4259	3,3559	-0,2879	-5,1814	-45,5900
2	0,3450	4,3779	-0,6451	-6,7594	-45,5900
1	0,2588	5,4789	-2,5737	-8,4593	-45,5900



```
if __name__ == "__main__":
    params = {
        'dim_bloco': (0.4, 0.27, 0.23),
        'l_ins': 0.04,
        'k_ins': 0.035,
        'h_int': 8.0,
        'h_ext': 22.0,
        'T_amb': 35.0,
        'n_TEC': 1,
        'alpha0': 0.050,
        'R0': 2.0,
        'K0': 0.8,
        'beta_alpha': -0.0015,
        'beta_R': 0.004,
        'beta_K': 0.0015,
        'cooling_type': 'water',
        'R_hs': 0.25,
        'm_dot_w': 0.05,
        'cp_w': 4180.0,
        'T_w_in': 35.0,
        'R_block': 0.02
    }
    sys = ParallelThermoelectricSensitivity(params)
```

Pior Cenário (Ar Parado/Sombra sem vento) (Sistema de Refrigeração a Ar)									
I	Tc	Th	Qc	Qh	Pel	COP	Alpha	R	K
6	25,5954	74,7079	5,8482	99,2698	93,4216	0,0626	0,0481	2,2012	0,8302
5	20,5927	64,2257	8,9591	73,0644	64,1052	0,1398	0,0487	2,1393	0,8209
4	18,3944	55,4526	10,3262	51,1314	40,8053	0,2531	0,0491	2,0954	0,8143
3	18,7892	48,2205	10,0806	33,0514	22,9707	0,4388	0,0494	2,0680	0,8102
2	21,6776	42,4252	8,2845	18,5630	10,2785	0,8060	0,0495	2,0564	0,8085
1	27,0544	38,0173	4,9410	7,5432	2,6022	1,8987	0,0494	2,0603	0,8090

I	dTh/dI	dQc/dI	dCOP/dI	dTc/dI	dTc/dL_ins
6	11,4111	-4,0474	-0,0652	6,5086	-39,8800
5	9,5938	-2,2105	-0,0916	3,5548	-39,8800
4	7,9803	-0,5451	-0,1405	0,8766	-39,8800
3	6,5011	1,0260	-0,2462	-1,6499	-39,8800
2	5,0978	2,5658	-0,5486	-4,1261	-39,8800
1	3,7175	4,1296	-2,1811	-6,6409	-39,8800

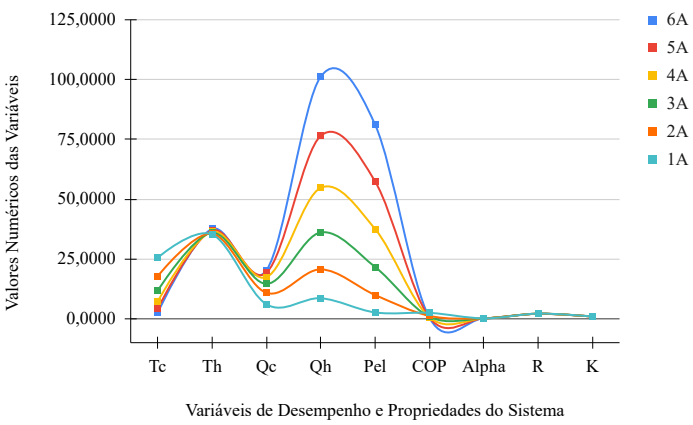


```
if __name__ == "__main__":
    params = {
        'dim_bloco': (0.4, 0.27, 0.23),
        'l_ins': 0.04,
        'k_ins': 0.035,
        'h_int': 8.0,
        'h_ext': 22.0,
        'T_amb': 35.0,
        'n_TEC': 1,
        'alpha0': 0.050,
        'R0': 2.0,
        'K0': 0.8,
        'beta_alpha': -0.0015,
        'beta_R': 0.004,
        'beta_K': 0.0015,
        'cooling_type': 'air',
        'R_hs': 0.25,
        'm_dot_w': 0.05,
        'cp_w': 4180.0,
        'T_w_in': 35.0,
        'R_block': 0.02
    }
    sys = ParallelThermoelectricSensitivity(params)
```

Pior Cenário (Ar Parado/Sombra sem vento) (Sistema de Refrigeração a Watercooler)									
I	Tc	Th	Qc	Qh	Pel	COP	Alpha	R	K
6	2,4850	37,5123	20,2194	101,3660	81,1466	0,2492	0,0504	1,9600	0,7940
5	4,1291	36,8972	19,1970	76,5468	57,3498	0,3347	0,0503	1,9641	0,7946
4	7,1694	36,3572	17,3064	54,7580	37,4516	0,4621	0,0502	1,9741	0,7961
3	11,6742	35,8937	14,5051	36,0571	21,5520	0,6730	0,0501	1,9903	0,7985
2	17,7342	35,5096	10,7367	20,5619	9,8251	1,0928	0,0499	2,0130	0,8019
1	25,4633	35,2096	5,9304	8,4565	2,5261	2,3476	0,0496	2,0427	0,8064

I	dTh/dI	dQc/dI	dCOP/dI	dTc/dI	dTc/dL_ins
6	0,6526	0,5991	-0,0716	-0,9634	-41,7900
5	0,5777	1,4505	-0,1023	-2,3326	-41,7900
4	0,5021	2,3377	-0,1587	-3,7593	-41,7900
3	0,4244	3,2742	-0,2805	-5,2653	-41,7900
2	0,3429	4,2743	-0,6285	-6,8736	-41,7900
1	0,2560	5,3527	-2,5074	-8,6078	-41,7900

```
if __name__ == "__main__":
    params = {
        'dim_bloco': (0.4, 0.27, 0.23),
        'l_ins': 0.04,
        'k_ins': 0.035,
        'h_int': 8.0,
        'h_ext': 22.0,
        'T_amb': 35.0,
        'n_TEC': 1,
        'alpha0': 0.050,
        'R0': 2.0,
        'K0': 0.8,
        'beta_alpha': -0.0015,
        'beta_R': 0.004,
        'beta_K': 0.0015,
        'cooling_type': 'water',
        'R_hs': 0.25,
        'm_dot_w': 0.05,
        'cp_w': 4180.0,
        'T_w_in': 35.0,
        'R_block': 0.02
    }
    sys = ParallelThermoelectricSensitivity(params)
```



Valor dos parâmetros utilizados para comparação de cenário entre Moraes e Kronbauer	
Moraes	Kronbauer
<pre>if __name__ == "__main__": params = { 'dim_bloco': (0.19, 0.11, 0.14), 'l_ins': 0.020, 'k_ins': 0.035, 'h_int': 8.0, 'h_ext': 22.0, 'T_amb': 28.2, 'n_TEC': 1, 'alpha0': 0.050, 'R0': 2.0, 'K0': 0.8, 'beta_alpha': -0.0015, 'beta_R': 0.004, 'beta_K': 0.0015, 'cooling_type': 'air', 'R_hs': 0.32164, 'm_dot_w': 0.05, 'cp_w': 4180.0, 'T_w_in': 28.2, 'R_block': 0.02, 'T_init': 28.2, 'stabilization_tolerance': 0.5, 'stabilization_rate_tolerance': 0.01, 'stabilization_max_time': 7200.0 } sys = ParallelThermoelectricSensitivity(params) I_test = 5.9</pre>	<pre>if __name__ == "__main__": params = { 'dim_bloco': (0.19, 0.11, 0.14), 'l_ins': 0.015, 'k_ins': 0.035, 'h_int': 8.0, 'h_ext': 22.0, 'T_amb': 28, 'n_TEC': 1, 'alpha0': 0.050, 'R0': 2.0, 'K0': 0.8, 'beta_alpha': -0.0015, 'beta_R': 0.004, 'beta_K': 0.0015, 'cooling_type': 'air', 'R_hs': 0.4184, 'm_dot_w': 0.05, 'cp_w': 4180.0, 'T_w_in': 28.2, 'R_block': 0.02, 'T_init': 28.2, 'stabilization_tolerance': 0.5, 'stabilization_rate_tolerance': 0.01, 'stabilization_max_time': 7200.0 } sys = ParallelThermoelectricSensitivity(params) I_test = 5</pre>